

Class Imbalance for Image Classification

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I. INTRODUCTION

In this assignment we study the problem of class imbalance in image classification tasks. The experiments are conducted using the Scenes15 dataset.

We simulate different imbalance scenarios in the training data and evaluate how the classifier behaves under these conditions. First, a baseline model is trained on the imbalanced dataset. Then, a weighted loss function is applied to reduce the effect of class imbalance.

The results show that weighted loss can improve the performance of minority classes, although the improvement varies depending on the imbalance structure.

II. METHOD

For Q1, one imbalanced version of the dataset is created using a subset of visually different classes from Scenes15. The severity of the imbalance is medium, which is 32 pictures per each minority class (kitchen, forest and coast). This experiment is used to compare the baseline model and the weighted loss strategy on a simpler classification problem.

For Q2, the experiments are extended to all 15 classes of Scenes15 using two imbalance structures. The first one is a *uniform minority* setting, where a small set of classes has fewer samples than the rest. The second one is a *long-tail* setting, where the number of samples decreases gradually across classes.

III. IMPLEMENTATION

The classifier used in all experiments is a simple convolutional neural network called SimpleCNN.

A. Introducing imbalance in the dataset

To simulate class imbalance, the imbalance profiles are generated using the `define_imbalance_profile` function from the `dlvsp-utils` library. This function determines the desired number of samples per class depending on the imbalance configuration. The final imbalanced training dataset is then created with the `apply_imbalance` function, while the test set remains unchanged.

B. Weighted loss for class imbalance

To address the imbalance problem, a weighted version of the cross entropy loss is used. The weights are computed based on the number of samples in each class, so that minority classes receive larger weights than majority classes.

These weights are passed to the `CrossEntropyLoss` function during training. As a result, classification errors for

minority classes contribute more to the total loss, encouraging the model to learn better representations for underrepresented classes.

IV. DATA

For all experiments the dataset Scene15 is used, but with modifications for each imbalance scenario that we are experimenting with:

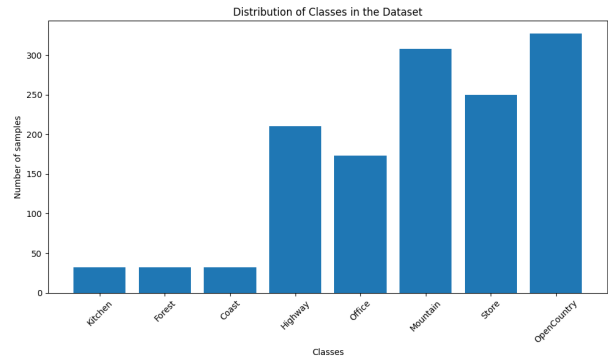


Fig. 1. Data distribution of the imbalanced training dataset - Q1

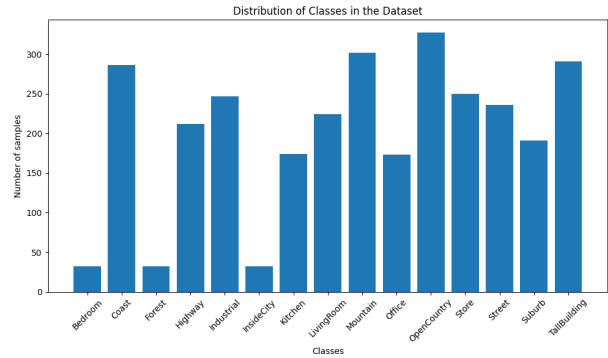


Fig. 2. Data distribution for uniform imbalance - Q2

V. RESULTS AND ANALYSIS

A. Question 1

Baseline

In the baseline experiment, where no strategy is used to address class imbalance, the minority classes obtain very low accuracy. The model tends to favor the majority classes and often misclassifies minority samples as one of the dominant categories, as we can see in the confusion matrix in Figure 4.

We can also see this in the per-class accuracies:

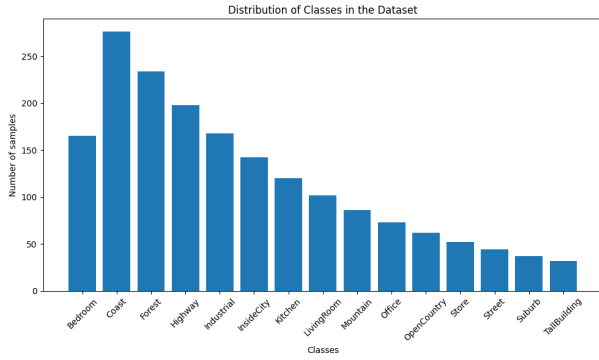


Fig. 3. Data distribution for long tail scenario - Q2

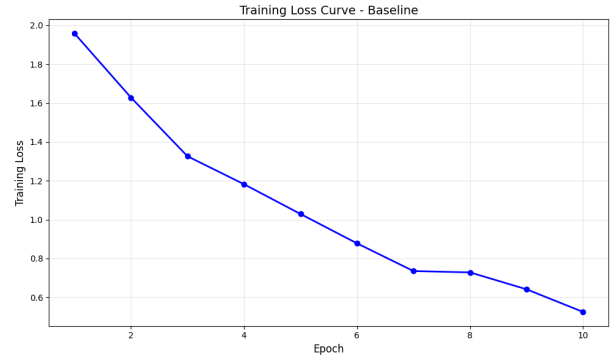


Fig. 5. Loss function for baseline

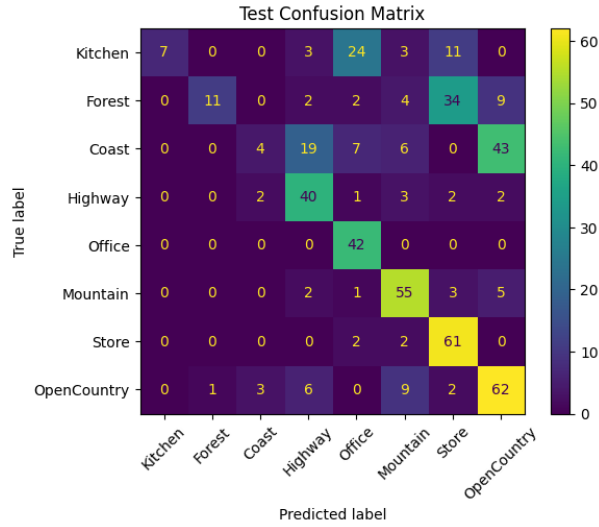


Fig. 4. Confusion matrix for baseline

- Kitchen: 14.58%
- Forest: 17.74%
- Coast: 5.06%
- Highway: 80.00%
- Office: 100.00%
- Mountain: 83.33%
- Store: 93.85%
- OpenCountry: 74.70%

Adding weighted loss strategy

We analyse the per-class accuracy for the weighted loss scenario as well:

- Kitchen: 70.83%
- Forest: 85.48%
- Coast: 56.96%
- Highway: 68.00%
- Office: 88.10%
- Mountain: 62.12%
- Store: 75.38%
- OpenCountry: 45.78%

We observe an improvement in the per-class accuracy when using the weighted loss strategy, although not all minority classes benefit equally. To further address class

difficulty, we experimented with increasing the weighting factors for harder classes. For example, a factor of 1.3 was applied to the Coast and OpenCountry classes.

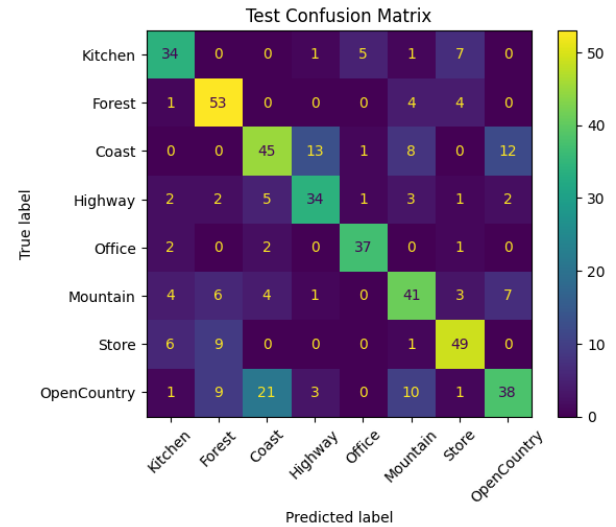


Fig. 6. Confusion matrix after weighted loss

Did the strategy improve minority classes compared to the baseline?

Yes, we conclude that the weighted loss strategy clearly improves the performance of the minority classes compared to the baseline. For example, the accuracy of the Kitchen class increases from 14.58% to 70.83%, Forest from 17.74% to 85.48%, and Coast from 5.06% to 56.96%. This shows that the model becomes better at recognizing underrepresented classes.

However, this improvement comes with a trade-off. Some majority classes experience a decrease in accuracy, such as OpenCountry, which drops from 74.70% to 45.78%. This happens because the weighted loss shifts the focus of the model toward minority classes.

B. Question 2

Scenario A

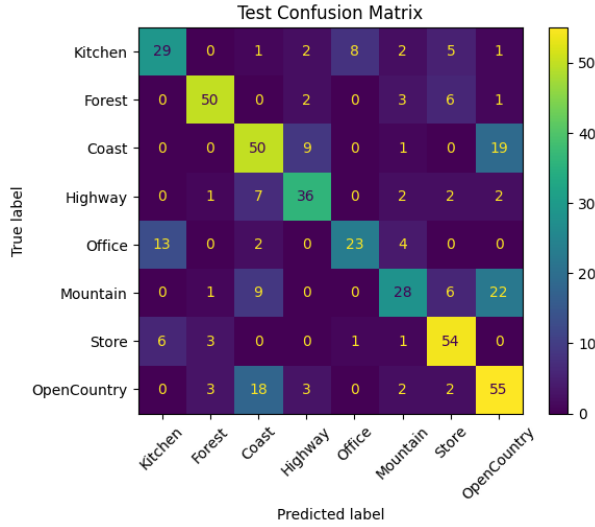


Fig. 7. Confusion matrix after weighted loss and factoring

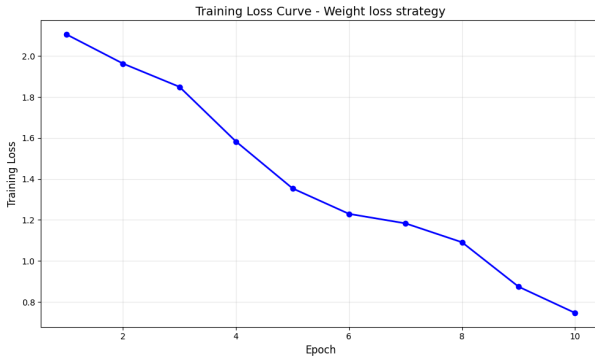


Fig. 8. Loss function for weighted loss

In this experiment we use all 15 classes from the Scenes15 dataset. It is similar to the one performed in Question 1, but now the model is trained and evaluated on the full dataset. The class distribution in this scenario was shown in Figure 2. We analyse the per-class accuracies in Table 1.

Scenario B

In this experiment we also use all 15 classes from the Scenes15 dataset, but the imbalance follows a long-tail distribution. In this case, the number of samples gradually decreases across classes, meaning that some classes contain significantly fewer training examples than others. We analyse the per-class accuracies in Table 2.

Comparison

The classes that benefit the most from the weighted loss strategy are generally the minority classes, since the method shifts the training focus toward underrepresented categories. However, the improvement also depends on the difficulty of each class, as visually distinctive classes tend to show larger accuracy gains.

In Scenario A, the imbalance is essentially binary (minority vs. majority). The weighted loss therefore focuses on a small set of clearly underrepresented classes, such as Bed-

TABLE I
PER-CLASS ACCURACY FOR SCENARIO A (BASELINE VS WEIGHTED LOSS)

Class	Baseline (%)	Weighted Loss (%)
Bedroom	1.96	31.37
Coast	83.78	83.78
Forest	43.84	78.08
Highway	62.50	66.67
Industrial	21.88	21.88
InsideCity	0.00	26.42
Kitchen	55.56	69.44
LivingRoom	44.62	32.31
Mountain	77.78	55.56
Office	54.76	69.05
OpenCountry	63.86	68.67
Store	93.85	70.77
Street	73.21	60.71
Suburb	86.00	84.00
TallBuilding	93.85	58.46

TABLE II
PER-CLASS ACCURACY FOR SCENARIO B (BASELINE VS WEIGHTED LOSS)

Class	Baseline (%)	Weighted Loss (%)
Bedroom	41.18	25.49
Coast	94.59	68.92
Forest	86.30	63.01
Highway	52.08	70.83
Industrial	42.19	17.19
InsideCity	66.04	20.75
Kitchen	72.22	47.22
LivingRoom	27.69	49.23
Mountain	25.00	15.28
Office	83.33	59.52
OpenCountry	22.89	71.08
Store	26.15	58.46
Street	44.64	55.36
Suburb	20.00	50.00
TallBuilding	4.62	44.62

room and Forest, which leads to significant improvements in their accuracy.

In Scenario B, the imbalance follows a long-tail distribution, where the focus must be spread across many classes at once. Instead of correcting a small group of minority classes, the model must compensate for a continuous range of class frequencies. As a result, the improvements for rare classes are often smaller and less stable.

Some tail classes, such as TallBuilding and Suburb, still improve significantly because they move from very low recognition to moderate accuracy. However, several middle-frequency classes suffer from increased confusion, and some majority classes such as Coast and Forest show a drop in performance.

Overall, long-tailed imbalance is harder to handle because many classes are underrepresented at the same time. The model must correct a continuous hierarchy of class frequencies rather than focusing on a small number of clearly defined minority classes.

A solution for long-tailed datasets

One possible solution for long-tailed datasets is proposed

in [1]. The authors introduce a Dynamic Label Space Adjustment (DLSA) method that focuses on balancing the label space instead of only modifying the data distribution or the loss function. The method progressively separates head and tail classes to reduce the bias toward majority classes and improve recognition of minority classes. We attempted to implement this approach, but the results are not yet satisfactory.

VI. CONCLUSIONS

The experiments showed that a baseline model trained on imbalanced data tends to favor majority classes and performs poorly on minority categories.

Applying a weighted loss strategy significantly improves the accuracy of minority classes, although this often leads to a slight decrease in performance for some majority classes due to attention loss.

Finally, the comparison between uniform imbalance and long-tail imbalance shows that long-tail distributions are harder to handle because many classes are underrepresented at the same time. In contrast, uniform imbalance separates classes into minority and majority groups, allowing the model to focus on correcting only a few minority classes.

VII. TIME LOG

- Andrea
 - Study: 1h
 - Coding: 1h
 - Writing: 1h
 - Testing: 2h
- Alexandra
 - Study: 1h
 - Coding: 1h
 - Writing: 3h
 - Testing: 0h
- Adham
 - Study: 1h
 - Coding: 2h
 - Writing: 2h
 - Testing: 2h factors and DLSA solution using AI

REFERENCES

- [1] Y. Xu, Y.-L. Li, J. Li, and C. Lu, “Constructing Balance from Imbalance for Long-tailed Image Recognition,” in *European Conference on Computer Vision (ECCV)*, 2022.